

Fun Maths

8:35 pm

Google

Q $\Rightarrow$ Majority ele

Given an array of size n (+ve no)
Return, if any no. exist where

$$\text{freq} > N/2$$

A: 1, 6, 1, 1, 2, 1

$$N = 6$$

$$6/2$$

$$\textcircled{1} \Rightarrow 4$$

Majority Ele

Quiz 1: 3, 4, 3, 6, 1, 3, 2, 5, 3, 3, 3

$$3 \Rightarrow 6$$

$$N = 11$$

$$11/2 \Rightarrow 5$$

Quiz 2: 4, 6, 5, 3, 4, 5, 6, 4, 4, 4

o/p: -1

$$N = 10$$

$$10/2 \Rightarrow 5$$

solution

① For every element get the freq
if $\text{freq} > N/2$

for ($i = 0 \rightarrow N-1$)

{ freq = 0;

for ($j = 0 \rightarrow N-1$)

✓ ~ ✓

$$freq_{++j}$$

```

if (freq > N/2)
    return a[i]

```

Solu 2:

8

1, 2, 3, 3, 3, 3, 3, 3, 4, 5, 6
6 → break

Solu 3:

SC: $O(N)$

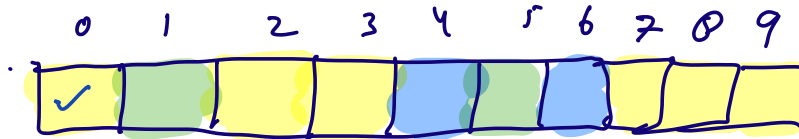
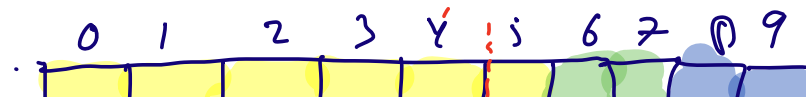
$O(n \log n)$

→ (3, 2) ←

⇕

Soln^y:

$O(N)$: TC

$$O(1) : SC$$
 $n/2$ 



→ Only one majority element can be there in an array

→ Count of ME will be greater than the combined count of remaining elements.

shivank IAS
— (i)

✓ Orange
Mahagatbandhan
✓
[Red
Green
Blue]



N	0
12	7
10	6
8	5
6	4

→ If we remove 2 distinct element from the array, ME will stay the same

3, 4, 3, 6, 1, 3, 2, 5, 3, 3, 3

$$N = 11$$

ME : ~~6~~ ~~6~~ ~~6~~, ~~8~~ ✓ 1 ✓

COUNT : 1 2 1 ~~0~~ 1 0 1 2 3 4 5 4 3 2 1 0 1

$N = 13$

$S \rightarrow 9$

~~6~~, ~~6~~, 2, 3

ele = arr[0], freq = 1

i = 1; i < N; i++

→ if (freq == 0) { ele = arr[i],
freq = 1; }

else if (ele != arr[i])

freq--;

→ else

freq++;

→ int c = 0;

→ i = 0; i < N; i++

if (ele == arr[i])

c++;

if (c > N/2) return ele,

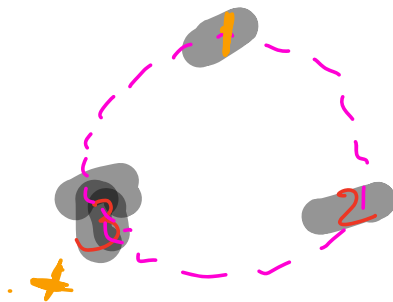
Break: 10:05

1, 2, 3, 3, 3, 3, 3

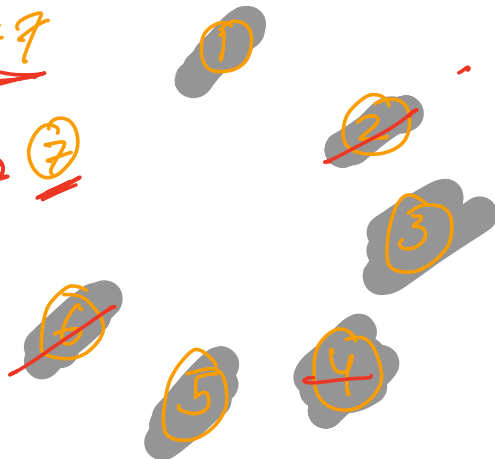
MG : 1 3

C : 1 2

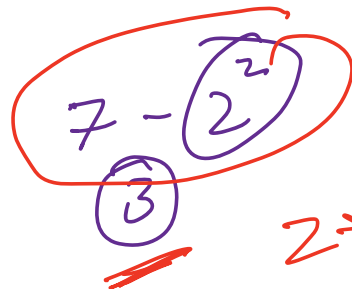
Joseph (Jewish) $\curvearrowright N$



$N=7$
~~X~~
~~7~~

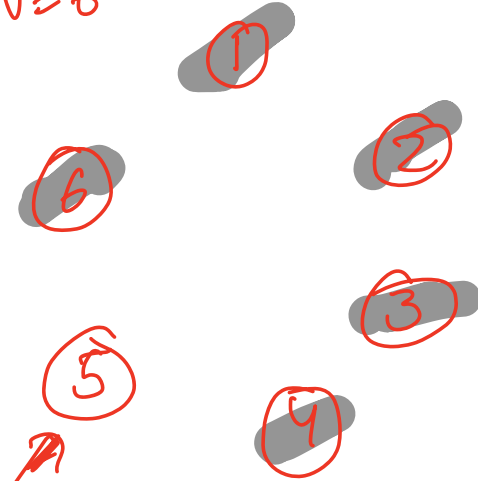


odd : N
even : $N-1$

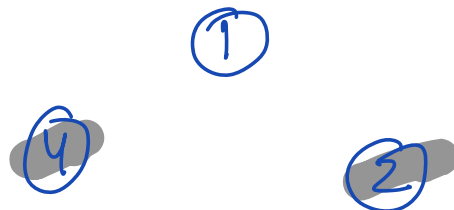


$2 \times 3 + 1$

$N=6$



$N=4$



③

2^0 ①

2' ①

2

⑧

①

7

②

⑥

③

5

4

$$N = 16$$

2*5H

16 9

①

② ✓



9



⑥

7



9

10

11

12

13

14

18

16

If N is a power of 2
we can say start elem
will be saved

$$N=11$$

$$11 - 2^3 = 3$$

$$2^3$$

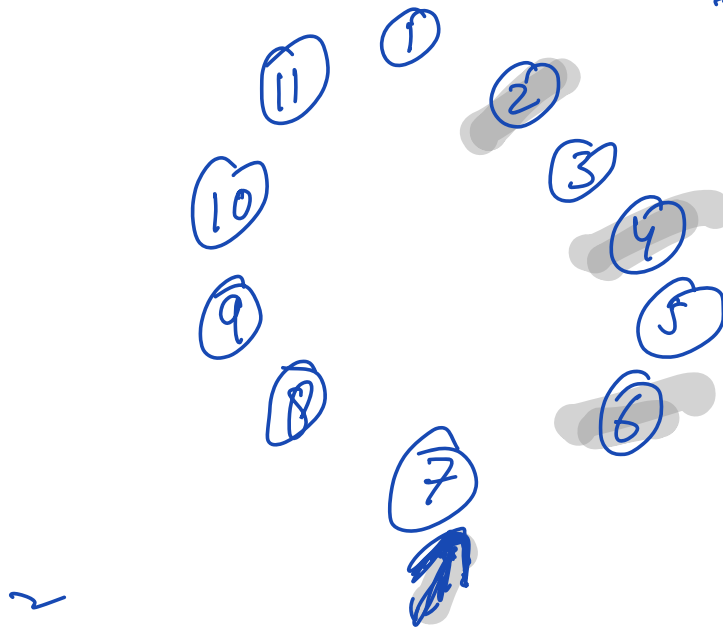
$$2 * 3 + 1$$

$$13 - 2^3 = 5$$

$$13 - 2^3 = 5$$

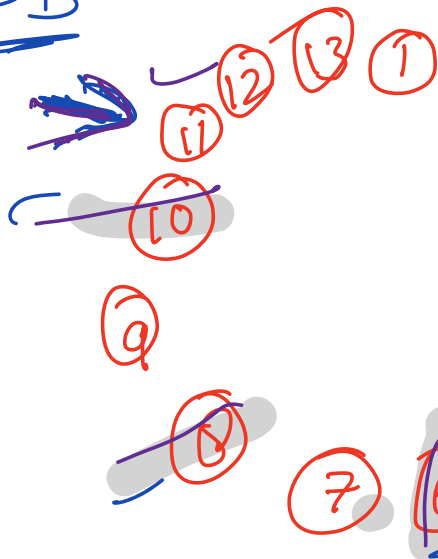
$$2 * 5 + 1 \Rightarrow 11$$

$$13 - 5$$



2

$$N=13$$

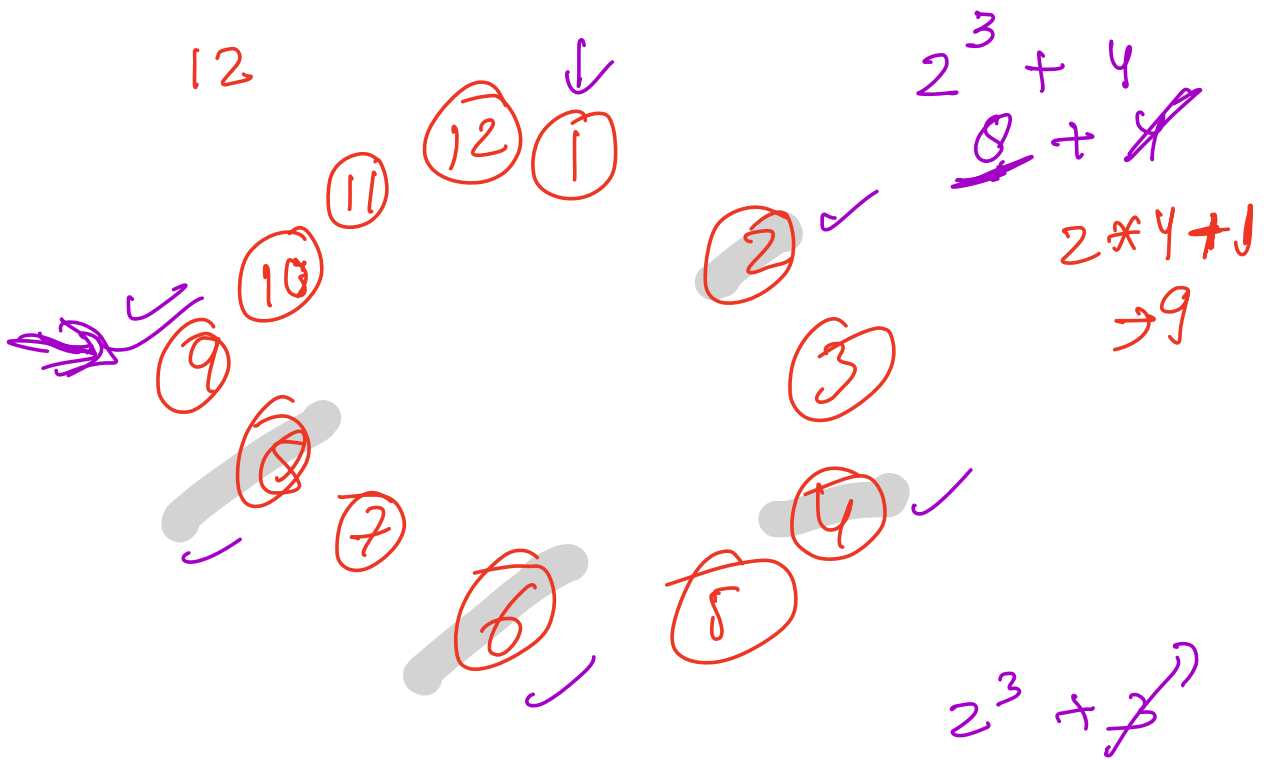


alive (N)

```

    p = (power of 2 <= N)
    return 2 * (N - p) + 1

```



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1, 2, 3, 3, 3, 3, 3

$$13 - 2^x$$

$$x = 3$$

$$\text{val} = 13 - 2^3$$

$$\text{val} = 5 \swarrow$$

$$\text{return } \underline{2\text{val}} + 1$$